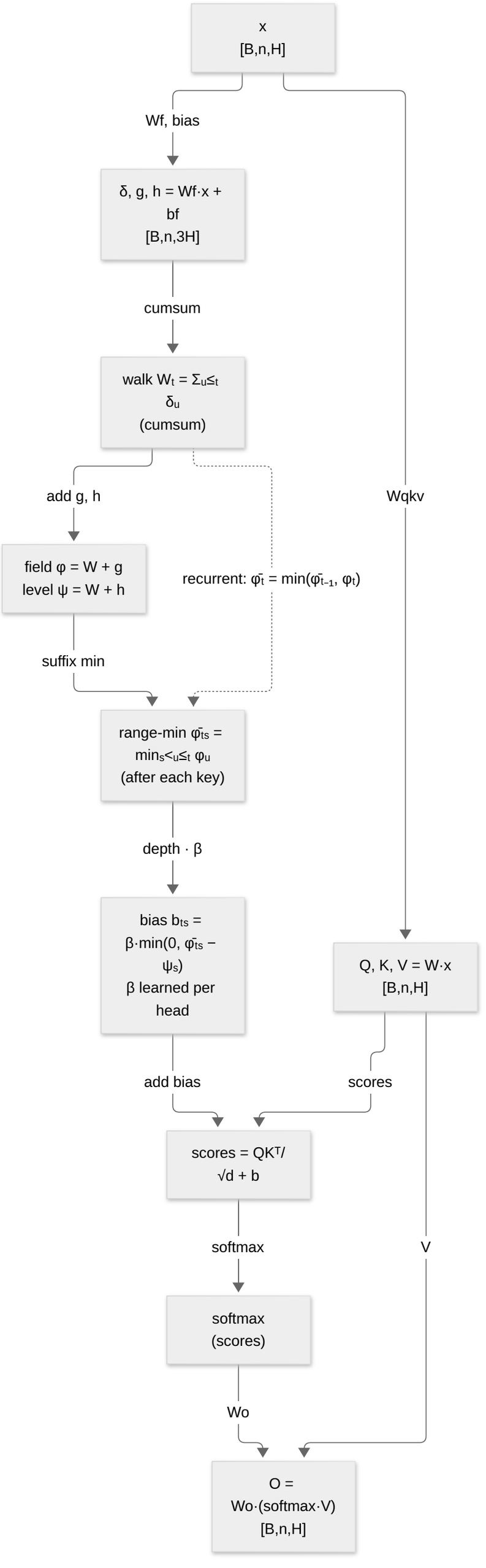

x
[B,n,H]
Wf, bias
δ, g, h = Wf·x + bf
[B,n,3H]
cumsum
walk $W_t = \Sigma_{u \le t} \delta_u$
(cumsum)
add g, h
Wqkv
field φ = W + g
level ψ = W + h
recurrent: $\bar{\varphi}_t = \min(\bar{\varphi}_{t-1}, \varphi_t)$
suffix min
range-min $\bar{\varphi}_{ts} = \min_{s<u\le t} \varphi_u$
(after each key)
depth · β
bias $b_{ts} = \beta \cdot \min(0, \bar{\varphi}_{ts} - \psi_s)$
β learned per head
Q, K, V = W·x
[B,n,H]
add bias
scores
scores = $QK^T/\sqrt{d} + b$
softmax
V
softmax
(scores)
Wo
O = Wo·(softmax·V)
[B,n,H]